Question 1: One-dimensional FDTD simulation

Develop a one-dimensional finite-difference time-domain (FDTD) solver based on Maxwell's equations to study electromagnetic-wave propagation in free space and interaction with a dielectric interface.

Instructions and expected outcomes

Parts A, B, and C are compulsory. On completing this assignment, a student should be able to

- formulate and implement a stable one-dimensional FDTD scheme directly from Maxwell's curl equations;
- extract propagation velocity, intrinsic impedance, reflection and transmission coefficients, and electromagnetic power flow from numerical field samples;
- compare the simulated results quantitatively with electromagnetic theory and explain any numerical discrepancy; and
- generate and interpret polarization trajectories from two transverse electric-field components.

Expected submission. Submit the source code, numerical formulation and parameters, clearly labelled plots or animations, the procedures used to extract physical quantities, comparisons with theory, and concise answers to all questions. State every additional assumption and parameter introduced in the simulation.

Evaluation and mark distribution. The regular assignment marks are divided equally among the three compulsory parts:

Part A	Part B	Part C
one-third ($33\frac{1}{3}\%$)	one-third ($33\frac{1}{3}\%$)	one-third ($33\frac{1}{3}\%$)

Part D is optional and carries no regular assignment marks. Grace credit will be considered **if and only if** Part D is executed exceptionally well, with correct physics, careful numerical validation, and clear analysis. Merely attempting or completing Part D does not earn grace credit. Any grace credit awarded may be used, at the instructor's discretion, during final grading when considering a possible grade change.

Simulation parameters

- Computational domain: $-3 \leq z \leq 3$ m.
- Initial medium: free space.
- Source location: $z = 0$.
- Excitation:

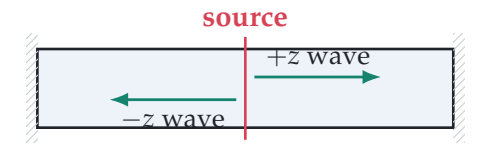
$$E_y(0,t) = \exp\left[-\left(\frac{t-t_0}{T}\right)^2\right] \sin(2\pi f t)\ \text{V/m},$$

  where

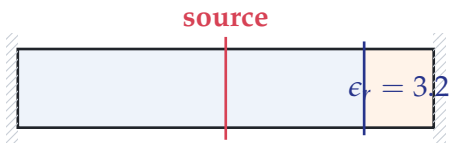
$$f = 1\ \text{GHz}, \qquad t_0 = 5\ \text{ns}, \qquad T = 2\ \text{ns}.$$

- Boundary condition: perfect electric conductor (PEC) at both ends of the computational domain.

Source and measurement note. The source at $z = 0$ launches waves in both the $+z$ and $-z$ directions. For dielectric-interface measurements, use the right-propagating wave as the incident wave. A reference free-space simulation may be used to separate its amplitude from the reflected field.



Part A: free-space domain with PEC ends



Part B: dielectric from $z = 2$ to 3 m

Part A: Propagation in free space

Implement a one-dimensional FDTD solver for Maxwell's equations.

1. Specify the spatial step Δz, time step Δt, total simulation time, spatial resolution, and Courant stability condition used in the implementation.
2. Animate the evolution of the electric and magnetic fields, or plot the field distributions at suitable time instants.
3. Estimate the propagation velocity of the electromagnetic pulse and compare it with the theoretical speed of light. Also estimate the intrinsic impedance of free space from the simulated electric and magnetic fields.
4. Explain why the reflected electric field undergoes a polarity reversal at a PEC boundary, whereas the reflected magnetic field does not.

Clearly state how the propagation velocity and field ratio are extracted from the numerical data. Measurements must be made before multiple PEC reflections overlap the pulse being measured.

Part B: Dielectric interface

Introduce a lossless dielectric occupying $2 \leq z \leq 3$ m, with

$$\epsilon_r = 3.2, \qquad \mu_r = 1.$$

1. Simulate the interaction of the incident pulse with the dielectric interface and identify the incident, reflected, and transmitted waves.
2. Measure the propagation velocity inside the dielectric and compare it with the theoretical value.
3. Estimate the electric-field reflection and transmission coefficients at the interface from the simulated amplitudes and compare them with the theoretical Fresnel coefficients at normal incidence.
4. Compute and plot the instantaneous Poynting vector

   $$\boldsymbol{S}(z,t) = \boldsymbol{E}(z,t) \times \boldsymbol{H}(z,t).$$

   Identify the direction of power flow for the incident, reflected, and transmitted waves. Use the time-integrated normal Poynting flux through suitable observation planes to verify energy conservation at the interface.

Important measurement window. The dielectric is terminated by a PEC at $z = 3$ m. Therefore, extract the first-interface reflection and transmission before the wave reflected from the right PEC returns to $z = 2$ m and overlaps them.

Part C: Polarization

Extend the FDTD formulation to include the transverse components E_x, E_y, H_x, and H_y. Use the smooth turn-on function

$$s(t) = 1 - \exp\left(-\frac{t}{T}\right), \qquad t \geq 0,$$

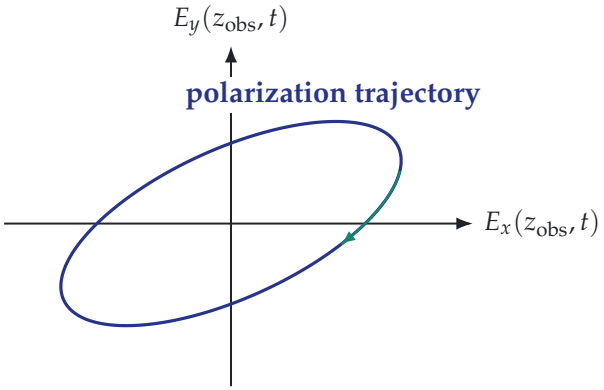
and excite

$$E_y(0,t) = s(t)\sin(2\pi f t)\ \mathrm{V/m},$$
$$E_x(0,t) = s(t)\sin(2\pi f t + \phi)\ \mathrm{V/m},$$

where

$$f = 1\ \mathrm{GHz}, \qquad T = 1\ \mathrm{ns}.$$

1. Select an observation point $z = z_{\mathrm{obs}}$ and plot $E_y(z_{\mathrm{obs}}, t)$ against $E_x(z_{\mathrm{obs}}, t)$ to obtain the polarization trajectory.
2. Sweep the phase difference ϕ and identify the conditions that produce linear, circular, and elliptical polarization.
3. Determine how the relative amplitudes and phase difference of E_x and E_y influence the polarization state. State the convention used to identify polarization handedness.



construct the locus from the simulated field samples at one point

Part D (optional): Bragg reflector

Using the FDTD solver developed in the previous parts, simulate a Bragg reflector made of alternating high- and low-refractive-index layers:

$$n_H = 3, \qquad n_L = 2,$$

with layer thicknesses

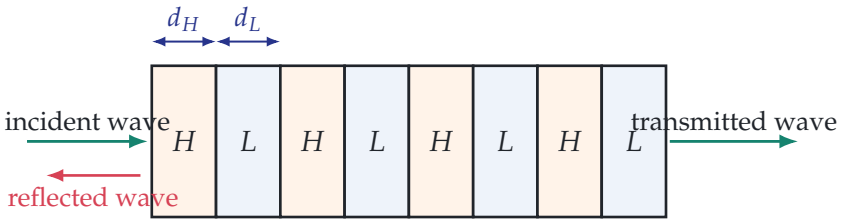
$$d_H = 2.5 \text{ cm}, \qquad d_L = 3.75 \text{ cm}.$$

Construct the periodic sequence

$$(H, L, H, L, H, L, \ldots)$$

and simulate it at

$$f = 1 \text{ GHz} \qquad \text{and} \qquad f = 4 \text{ GHz}.$$



Choose the position of the reflector and the number of (H, L) periods, N, and report both values. Provide sufficient free-space regions for source injection and for reflected/transmitted-field measurements. The spatial mesh must resolve the shortest wavelength present in the highest-index material.

Questions

1. Explain why $d_H = 2.5$ cm and $d_L = 3.75$ cm were selected. Determine the frequency at which strong Bragg reflection is expected.
2. Compare the reflected and transmitted waves at 1 GHz and 4 GHz. Which frequency experiences stronger reflection, and which experiences greater transmission?
3. Investigate what happens when the number of layers is increased. Comment on the peak reflectivity and the spectral response.